

ChainObserver 🔍

Paste a failed Ethereum tx → get the root cause **and** the fix — in **seconds**, not a 20-minute Etherscan dig

Powered by **Gemini 2.5 Flash** + custom Ethereum **MCP** tools

ETHGlobal Lisbon 2026 · live at johnlee007-chainobserver.hf.space

The problem

Every Ethereum dev has stared at this:

| `execution reverted` · `0xa1148100` · `TRANSFER_FROM_FAILED`

Diagnosing one failed transaction means:

- 5 min digging through **Etherscan**
- decoding a **4-byte custom error** by hand
- guessing whether it's allowance, slippage, balance, gas, or a contract bug

Multiply by every failed tx, every day. The answer is on-chain — you just don't have time to dig it out.

The solution

One input: a tx hash. One output: a structured diagnosis.

```
$ chainobserver 0xaa78010d...
```

- **Root cause** in plain English
- **Failure category** (allowance / slippage / balance / gas / revert)
- **The fix** — exactly what to do next
- **Confidence** + an Etherscan link

Works on **mainnet, Arbitrum, Base, Optimism, Polygon.**

How it works

An agent, not a script. Gemini 2.5 Flash plans its own investigation over 5 purpose-built **Ethereum MCP tools**:

`get_transaction_receipt` → `decode_revert_reason` → `get_contract_info` →
`simulate_transaction` → `get_pool_info`

- The model decides which tool to call next, and stops when confident (≤ 5 calls)
- Custom Solidity errors resolved via **4byte.directory** + **Sourcify**
- Reusable from any MCP client (Claude Desktop, Cursor) — the tools outlive the demo

Proof — 8 real mainnet failures, 6 categories

Failure type	Example	E2E time
insufficient_allowance	Uniswap Universal Router	18.6s
insufficient_balance	USDC transfer	11.9s
slippage_exceeded	DEX swap (<i>also CoW, ParaSwap</i>)	21.4s
contract_revert	Seaport <code>0xa1148100</code>	~36s
out_of_gas	gas at 99.5% of limit	tool-verified

8 / 8 correctly classified ✓ — and on the 4 end-to-end timed runs:

100% accuracy · 21.8s avg · 3.25 tool calls (target ≤5).

Decoded a Seaport custom error *and* a ParaSwap `QuoteExpired()` with **zero hardcoding**.

Why it matters

- **Real intelligence, not a lookup table** — the agent reasons over live chain data and handles errors it's never seen.
- **MCP-native** — the 5 Ethereum tools are a reusable building block for the whole agent ecosystem, not a one-off.
- **Multi-chain from day one** — same agent, five networks.
- **Live now** — public, no install: `johnlee007-chainobserver.hf.space`.

What's next

- Auto-suggest the corrected transaction (ready-to-send calldata)
- Wallet & block-explorer integration ("Diagnose" button on any failed tx)
- Expand the MCP toolset → L2-specific failure modes, MEV/sandwich detection
- Part of **observer-ai** — agentic diagnostics across platforms (see also: PipelineGuard for CI)

ChainObserver

Failed tx in. Root cause + fix out. In seconds.

Try it: johnlee007-chainobserver.hf.space

Code: github.com/observer-ai-org/chainobserver

Built with: Gemini 2.5 Flash · Ethereum MCP · Python

Thank you — questions?